CHAPTER SIX

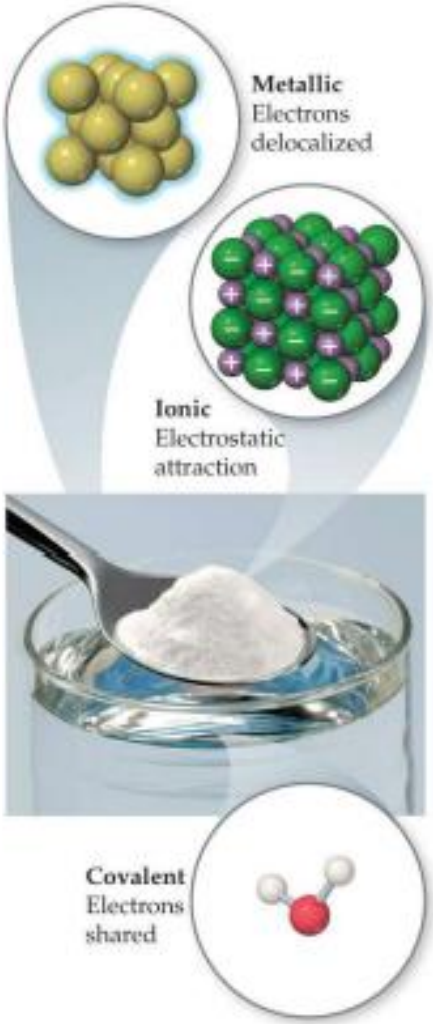
Chemical Bonding and Molecular Geometry







Introduction



Chemical Bonds

Three basic types of bonds:

- Ionic
  - Electrostatic attraction between ions
- Covalent
  - Sharing of electrons
- Metallic
  - Metal atoms bonded by free electrons

6.1 IONIC BOND

- Ions are atoms or molecules bearing an electrical charge.
- A cation (a positive ion) forms when a neutral atom **loses one or more electrons** from its valence shell,
- An anion (a negative ion) forms when a neutral atom **gains one or more electrons** in its valence shell.
- Ionic solids exhibit:
  - ✓ a crystalline structure and tend to be rigid and brittle;
  - ✓ they also tend to have high melting and boiling points, which suggests that ionic bonds are very strong.
- Ionic solids are
  - ✓ also poor conductors of electricity for the same reason-the strength of ionic bonds prevents ions from moving freely in the solid state.

6.1.1. The Formation of Ionic Compounds

- Binary ionic compounds are composed of just two elements: a metal (which forms the cations) and a nonmetal (which forms the anions).

  Example: NaCl is a binary ionic compound.

- Many metallic elements have relatively low ionization potentials and lose electrons easily.
  - ✓ These elements lie to the left in a period or near the bottom of a group on the periodic table.
- Nonmetal atoms have relatively high electron affinities and thus readily gain electrons lost by metal atoms, thereby filling their valence shells.
  - ✓ are found in the upper right corner of the periodic table.

Group	1A	2A	3A	4A	5A	6A	7A	8A
Element	Li	Be	B	C	N	O	F	Ne
Electron Configuration	$[He]2s^1$	$[He]2s^2$	$[He]2s^2 2p^1$	$[He]2s^2 2p^2$	$[He]2s^2 2p^3$	$[He]2s^2 2p^4$	$[He]2s^2 2p^5$	$[He]2s^2 2p^6$
Lewis Symbol	Li·	·Be·	·B·	·C·	·N:	:O:	:F:	:Ne:
	Na	Mg	Al	Si	P	S	Cl	Ar
	$[Ne]3s^1$	$[Ne]3s^2$	$[Ne]3s^2 3p^1$	$[Ne]3s^2 3p^2$	$[Ne]3s^2 3p^3$	$[Ne]3s^2 3p^4$	$[Ne]3s^2 3p^5$	$[Ne]3s^2 3p^6$
	Na·	·Mg·	·Al·	·Si·	·P:	:S:	:Cl:	:Ar:

6.1.2. Electronic Structures of Cations and Anions

- When forming a cation, an atom of a main group element tends to lose all of its valence electrons,

Example: groups 1 (the alkali metals) and 2 (the alkaline earth metals)

calcium is a group 2 element whose neutral atoms have 20 electrons and a ground state electron configuration of

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

When a Ca atom loses both of its valence electrons, the result is a cation with 18 electrons, a +2 charge, and an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^6$. The Ca^{2+} ion is therefore **isoelectronic with the noble gas Ar**.

- Transition and inner transition metal elements **behave differently** than main group elements.
  - ✓ Most transition metal cations have +2 or +3 charges that result from the loss of their outermost s electron(s) first, sometimes followed by the loss of one or two d electrons from the next-to-outermost shell.

For example, iron ($1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$) forms

the ion Fe^{2+} ($1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$) by the loss of the 4s electrons

the ion Fe^{3+} ($1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$) by the loss of the 4s electrons and one of the 3d electrons.

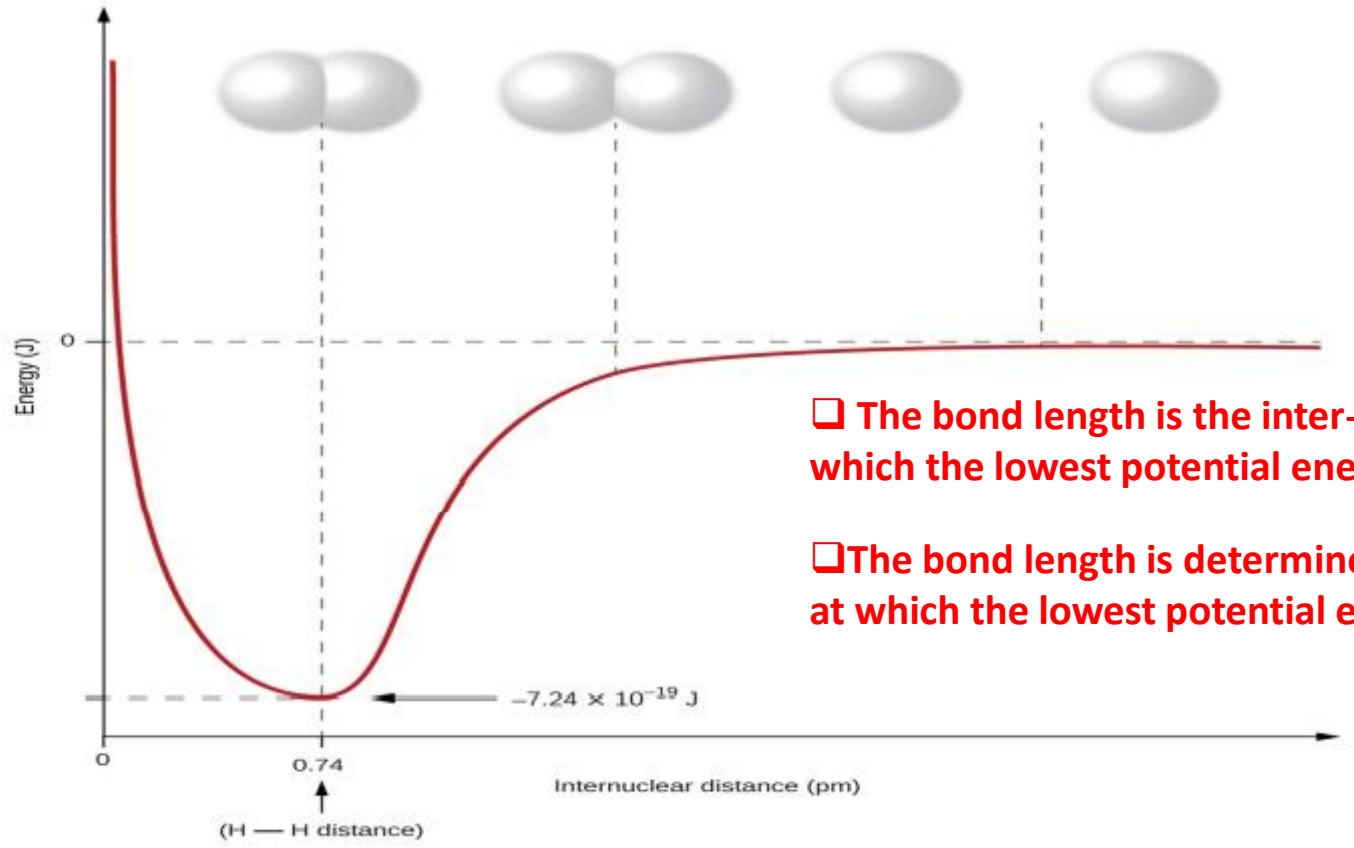
6.2 Covalent Bonding

- covalent bonds are forms of bonds resulted from the mutual attraction of atoms for a “shared” pair of electrons.
- Covalent bonds are formed between two atoms when both have similar tendencies to attract electrons to themselves (i.e., when both atoms have identical or fairly similar ionization energies and electron affinities).
- Compounds that contain covalent bonds exhibit different physical properties than ionic compounds.
  - The attraction between molecules, which are electrically neutral, is weaker than that between electrically charged ions, covalent compounds generally have much lower melting and boiling points than ionic compounds
  - Many covalent compounds are liquids or gases at room temperature, and, in their solid states, they are typically much softer than ionic solids.
  - Most covalent compounds are insoluble in water; since they are electrically neutral, they are poor conductors of electricity in any state.

6.2.1. Formation of Covalent Bonds

- Nonmetal atoms frequently form covalent bonds with other nonmetal atoms

Example: the hydrogen molecule, H_2



- **The bond length is the inter-nuclear distance at which the lowest potential energy is achieved.**
- **The bond length is determined by the distance at which the lowest potential energy is achieved.**

6.2.1. Formation of Covalent Bonds

- It is essential to remember:
  - ✓ that energy must be added to **break chemical bonds** (an endothermic process)
  - ✓ whereas **forming chemical bonds** releases energy (an exothermic process).
- In the case of H_2, the covalent bond is very strong; a large amount of energy, 436 kJ, must be added to break the bonds in one mole of hydrogen molecules and cause the atoms to separate:

$$H_2(g) \rightarrow 2H(g) \quad \Delta H = 436kJ$$

Conversely, the same amount of energy is released when one mole of H_2 molecules forms from two moles of H atoms:

$$2H(g) \rightarrow H_2(g) \Delta H = -436kJ$$

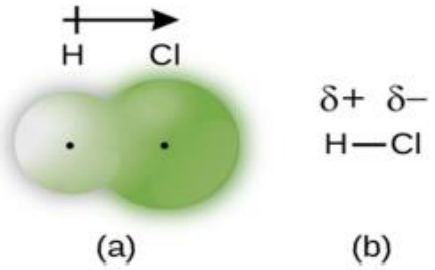
6.2.2. Polarity of Covalent Bonds

- If the atoms that form a covalent bond are identical, as in H_2, Cl_2, and other diatomic molecules, then the electrons in the bond must be **shared equally**.
  - ✓ In the case of Cl_2 each Cl shares one electron with the other, forming one covalent bond:
  - ✓ The total number of electrons around each individual atom consists of six nonbonding electrons and two shared

$$\mathbf{Cl + Cl \rightarrow Cl_2}$$

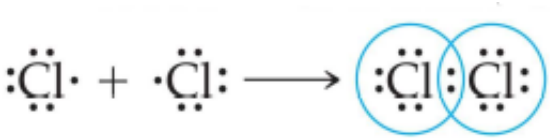
- When the atoms linked by a covalent bond are different, the bonding electrons are shared, but no longer equally.

For example, the electrons in the H–Cl bond

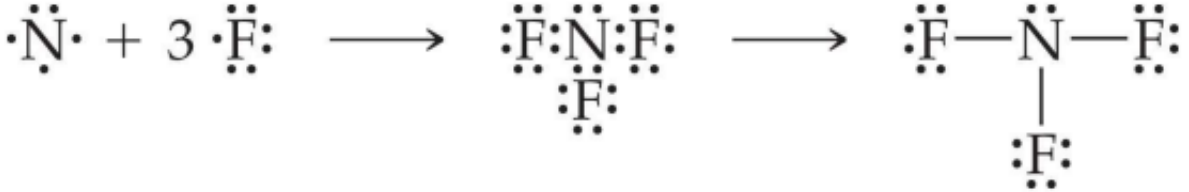


6.3. Lewis structures

- The Lewis structure of a chlorine molecule indicates that each Cl atom has **three pairs of electrons that are not used in bonding** (called lone pairs) and **one shared pair of electrons** (written between the atoms called bonding pair).
- A dash (or line) is sometimes used to indicate a shared pair of electrons:



Lewis structures are representations of molecules showing all electrons, bonding and nonbonding.







The Octet Rule

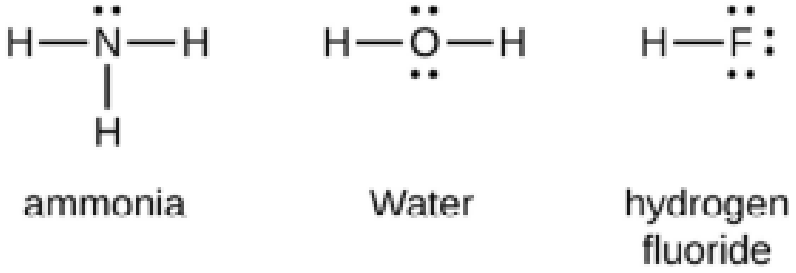
- The tendency of main group atoms to form enough bonds to **obtain eight valence electrons** is known as the octet rule.
- The number of bonds that an atom can form can often be predicted from the **number of electrons needed to reach an octet** (eight valence electrons);
- For example, each atom of a group 14 element (Group IVA) has four electrons in its outermost shell and therefore **requires four more electrons to reach an octet**.

Example: carbon in CCl_4 (carbon tetrachloride) and

silicon in SiH_4 (silane).

6.3. Lewis structures

- Group 15 elements (Group VA) such as nitrogen have **five valence electrons** in the atomic Lewis symbol: **one lone pair and three unpaired electrons**.
  - To obtain an octet, these atoms form **three covalent bonds**, as in NH_3 (ammonia).

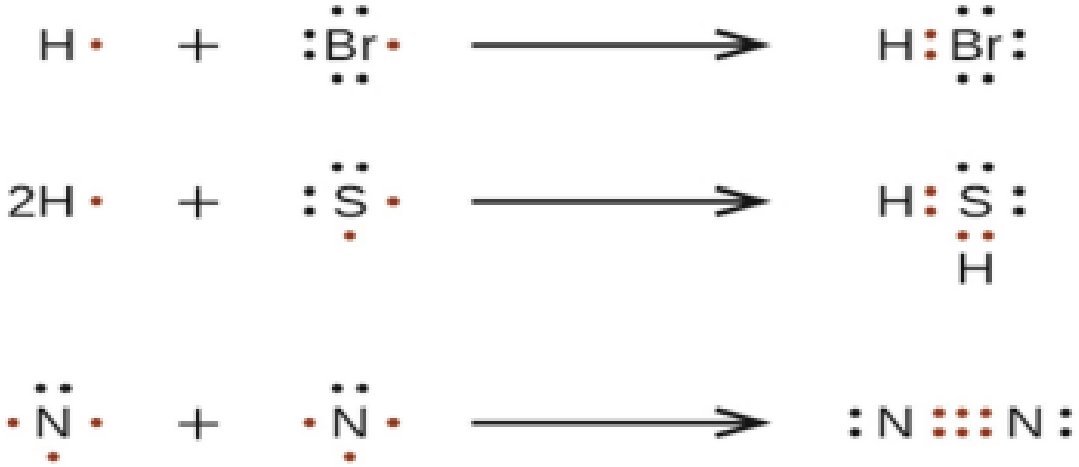


- Oxygen and other atoms in group 16 (Group VIA) obtain an octet by forming two covalent bonds:

6.3.1. Writing Lewis Structures with the Octet Rule

- For very simple molecules and molecular ions, we can write the Lewis structures by pairing up the unpaired electrons on the constituent atoms.

  See these examples:



6.3.1. Writing Lewis Structures with the Octet Rule

For more complicated molecules and molecular ions, it is helpful to follow the step-by-step procedure outlined here:

1. Determine the total number of valence (outer shell) electrons.
   - ✓ For cations, **subtract one electron** for each positive charge.
   - ✓ For anions, **add one electron** for each negative charge.
2. Draw a skeleton structure of the molecule or ion, arranging the atoms around a central atom.
   - ✓ Generally, the least electronegative element should be placed in the center.
   - ✓ Connect each atom to the central atom with a single bond (one electron pair)
3. Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen), completing an octet around each atom. Place all remaining electrons on the central atom.
4. Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

6.3.1. Writing Lewis Structures with the Octet Rule

- Let us determine the Lewis structures of SiH_4, CHO_2^- , NO^+ , and OF_2 as examples in following this procedure:

Step 1: Determine the total number of valence (outer shell) electrons in the molecule or ion. For a molecule, we add the number of valence electrons on each atom in the molecule: SiH_4

Si: 4 valence electrons/atom×1 atom = 4

+H: 1 valence electron/atom×4atoms = 4

= 8 valence electrons

For a negative ion, such as CHO_2^-, we add the number of valence electrons on the atoms to the number of negative charges on the ion (one electron is gained for each single negative charge):

CHO_2^-

C: 4 valence electrons/atom×1 atom=4

H: 1 valence electron/atom×1 atom=1

O: 6 valence electrons/atom×2 atoms=12

+1 additional electron

=18 valence electrons

6.3.1. Writing Lewis Structures with the Octet Rule

- ✓ For a positive ion, such as NO^+ , we add the number of valence electrons on the atoms in the ion and then subtract the number of positive charges on the ion (one electron is lost for each single positive charge) from the total number of valence electrons:

NO^+

N: 5 valence electrons/atom×1 atom=5

O: 6 valence electron/atom×1 atom=6

+ −1 electron (positive charge = -1

=10 valence electrons

Since OF_2 is a neutral molecule, we simply add the number of valence electrons:

OF_2

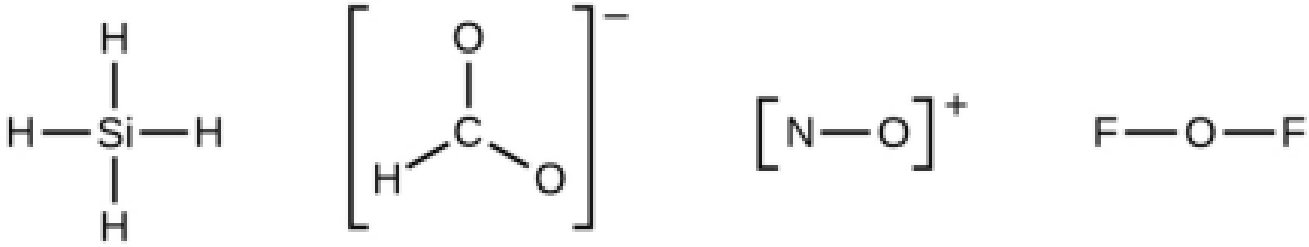
O: 6 valence electrons/atom×1 atom=6

F: 7 valence electrons/atom×2 atoms = 14

= 20 valence electrons

6.3.1. Writing Lewis Structures with the Octet Rule

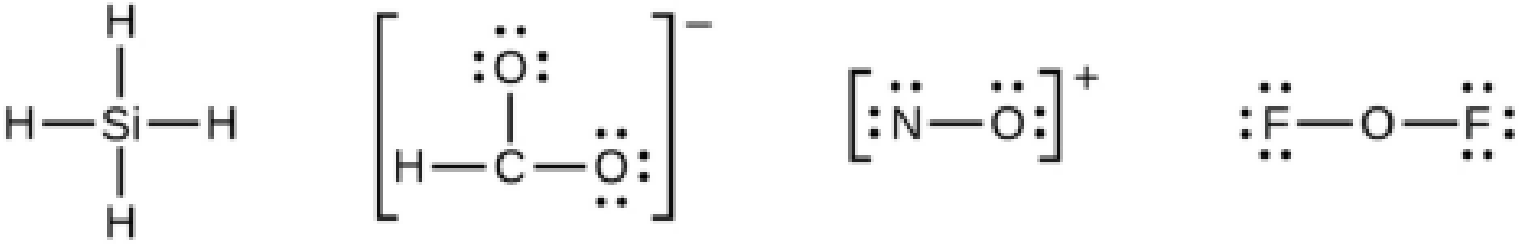
Step 2: Draw a skeleton structure of the molecule or ion, arranging the atoms around a central atom and connecting each atom to the central atom with a single (one electron pair) bond. (Note that we denote ions with brackets around the structure, indicating the charge outside the brackets:)



- ✓ In general, the less electronegative elements are more likely to be central atoms. In CHO_2^- , the less electronegative carbon atom occupies the central position with the oxygen and hydrogen atoms surrounding it.
- ➢ Other examples include **P** in $POCl_3$, **S** in SO_2, and **Cl** in ClO_4^-.
- ➢ An exception is that hydrogen is almost never a central atom.
- ➢ As the most electronegative element, fluorine also cannot be a central atom.

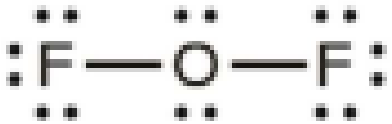
6.3.1. Writing Lewis Structures with the Octet Rule

Step 3: Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen) to complete their valence shells with an octet of electrons. There are no remaining electrons on SiH_4, so it is unchanged:



Step 4: Place all remaining electrons on the central atom.

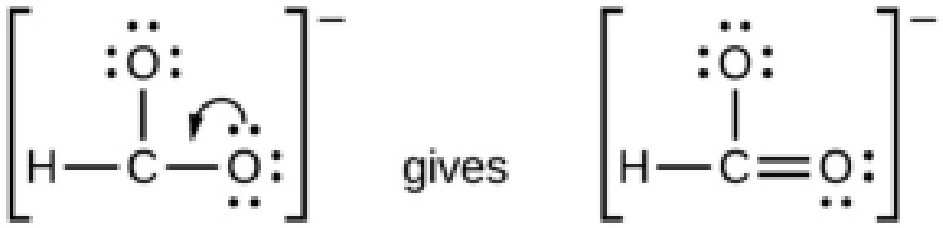
- ✓ For SiH_4, CHO_2^-, and NO^+ , there are no remaining electrons; we already placed all of the electrons determined in Step 1.
- ✓ For OF_2, we had 16 electrons remaining in Step 3, and we placed 12, leaving 4 to be placed on the central atom:



6.3.1. Writing Lewis Structures with the Octet Rule

Step 5: Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

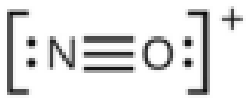
- ✓ SiH_4: Si **already has an octe**t, so nothing needs to be done.
- ✓ CHO_2^- : We have distributed the valence electrons as lone pairs on the oxygen atoms, but the **carbon atom lacks an octet**:



- ✓ NO^+ : For this ion, we added eight valence electrons, but neither atom has an octet. We cannot add any more electrons since we have already used the total that we found in Step 1, so we must **move electrons to form a multiple bond:**



- ✓ This still does not produce an octet, so we must move another pair, **forming a triple bond:**

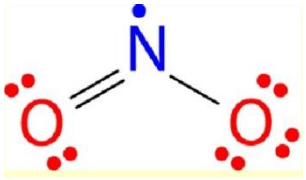


6.3.2. Exceptions to the Octet Rule

Many covalent molecules have central atoms that **do not have eight electrons in their Lewis structures**. These molecules fall into three categories:

- Odd-electron molecules have an odd number of valence electrons, and therefore have an unpaired electron.

Example:

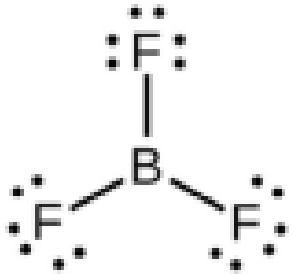


Nitrogen dioxide has 17 valance electrons

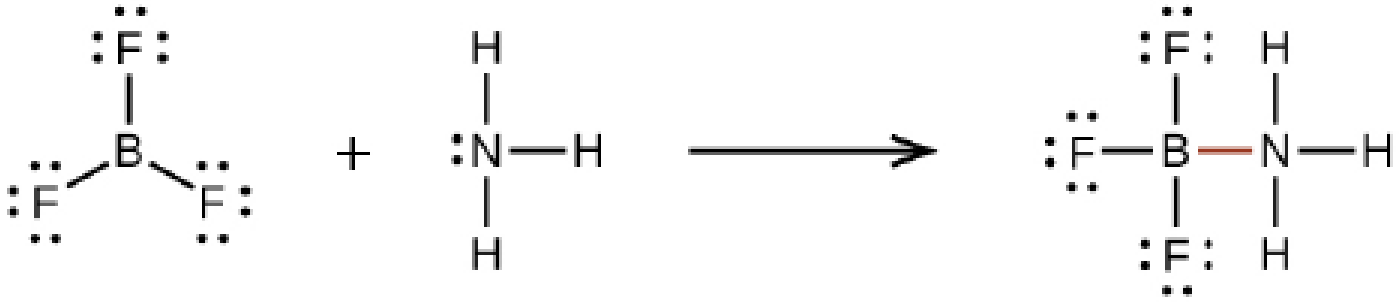
- Electron-deficient molecules have a central atom that has fewer electrons than needed for a noble gas configuration.

Example:





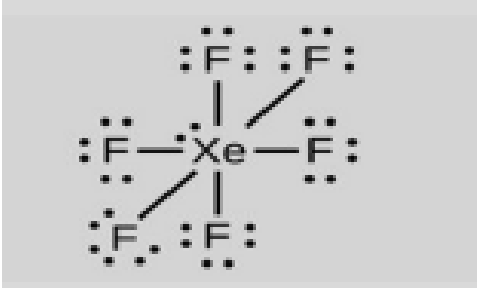
NB: An atom like the boron atom in BF_3, which does not have eight electrons, is very reactive. It readily combines with a molecule containing an atom with a lone pair of electrons. For example, NH_3 reacts with BF_3 because the lone pair on nitrogen can be shared with the boron atom



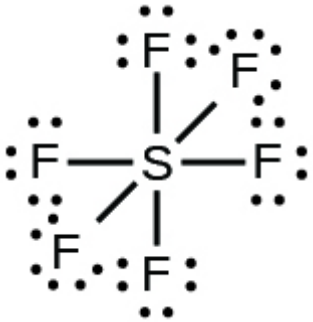
6.3.2. Exceptions to the Octet Rule

✓Hypervalent molecules have a central atom that has more electrons than needed for a noble gas configuration.

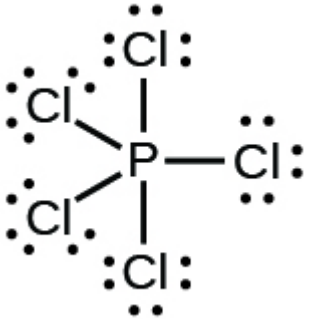
Example: Xenon hexafluoride



Sulfur hexafluoride



Phosphorus pentachloride



6.4.1. Calculating Formal Charge

- The formal charge of an atom in a molecule is the **hypothetical charge the atom** would have if we could redistribute the electrons in the bonds evenly between the atoms.

- The **sum of the formal charges of all atoms in a molecule must be zero**; the sum of the formal charges in an ion should **equal the charge of the ion**.

Formal Charge =

valance shell electrons –non-bonding elec. -1/2# bonding electrons

$$\text{Formal charge} = \left(\begin{array}{c}\text{number of valence}\\ \text{electrons in the}\\ \text{neutral atom}\end{array}\right) - \left(\begin{array}{c}\text{number of valence}\\ \text{electrons around the}\\ \text{atom in the molecule}\end{array}\right)$$

$$\text{Formal charge} = \left(\begin{array}{c}\text{number of valence}\\ \text{electrons in the}\\ \text{neutral atom}\end{array}\right) - \frac{1}{2}\left(\begin{array}{c}\text{number of electrons}\\ \text{in covalent bonds}\end{array}\right) - \left(\begin{array}{c}\text{number of electrons}\\ \text{in lone pairs}\end{array}\right)$$

Group number

remember that these electrons are shared by the atoms in the bond

OR

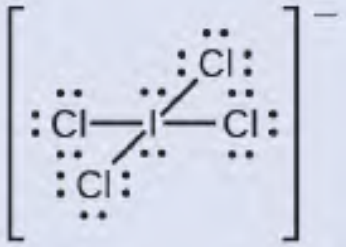
$$\text{Formal charge} = (\text{Group number}) - \left(\begin{array}{c}\text{number of}\\ \text{covalent bonds}\end{array}\right) - \left(\begin{array}{c}\text{number of electrons}\\ \text{in lone pairs}\end{array}\right)$$

Calculating Formal Charge from Lewis Structures

Assign formal charges to each atom in the interhalogen ion ICl_4^-.

Solution

Step 1. *We divide the bonding electron pairs equally for all I–Cl bonds:*



Step 2. *We assign lone pairs of electrons to their atoms.* Each Cl atom now has seven electrons assigned to it, and the I atom has eight.

Step 3. *Subtract this number from the number of valence electrons for the neutral atom:*

I: $7 - 8 = -1$

Cl: $7 - 7 = 0$

The sum of the formal charges of all the atoms equals −1, which is identical to the charge of the ion (−1).

Check Your Learning

Calculate the formal charge for each atom in the carbon monoxide molecule:



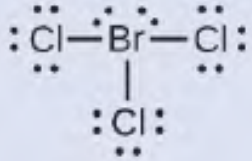
Answer: C −1, O +1

Calculating Formal Charge from Lewis Structures

Assign formal charges to each atom in the interhalogen molecule $BrCl_3$.

Solution

Step 1. *Assign one of the electrons in each Br–Cl bond to the Br atom and one to the Cl atom in that bond:*



Step 2. *Assign the lone pairs to their atom.* Now each Cl atom has seven electrons and the Br atom has seven electrons.

Step 3. *Subtract this number from the number of valence electrons for the neutral atom.* This gives the formal charge:

Br: $7 - 7 = 0$

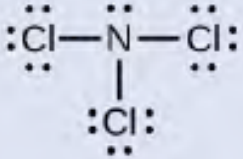
Cl: $7 - 7 = 0$

All atoms in $BrCl_3$ have a formal charge of zero, and the sum of the formal charges totals zero, as it must in a neutral molecule.

Check Your Learning

Determine the formal charge for each atom in NCl_3.

Answer: N: 0; all three Cl atoms: 0



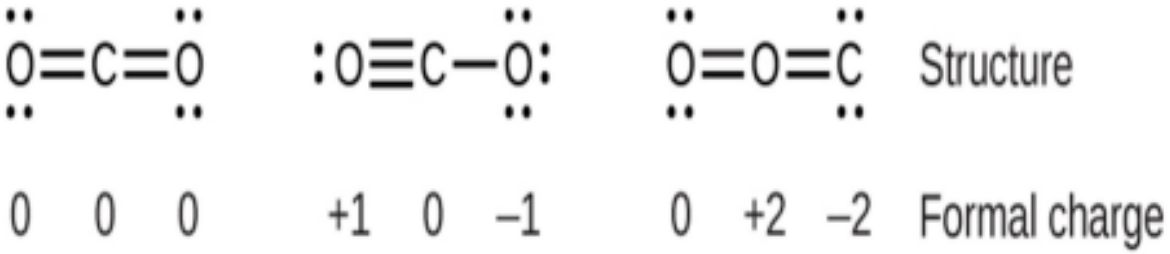
6.4.2. Predicting molecular structure using formal charge

- A few guidelines involving formal charge can be helpful in deciding which of the possible structures is most likely for a particular molecule or ion:

1. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero.
2. If the Lewis structure must have nonzero formal charges, the arrangement with the smallest nonzero formal charges is preferable.
3. Lewis structures are preferable when adjacent formal charges are zero or of the opposite sign.
4. When we must choose among several Lewis structures with similar distributions of formal charges, the structure with the negative formal charges on the more electronegative atoms is preferable.

6.4.2. Predicting molecular structure using formal charge

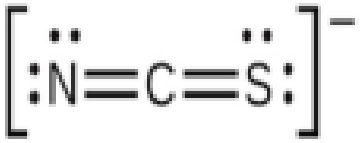
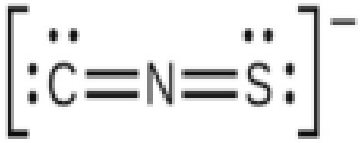
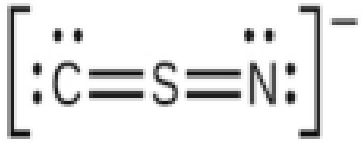
- To see how these guidelines apply, let us consider some possible structures for carbon dioxide, CO_2.



- Comparing the three formal charges, we can definitively identify the structure on the left as preferable because it has only formal charges of zero (Guideline 1).

6.4.2. Predicting molecular structure using formal charge

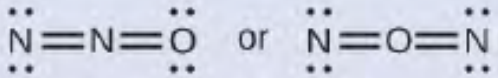
- As another example, the thiocyanate ion, an ion formed from a carbon atom, a nitrogen atom, and a sulfur atom, could have three different molecular structures: NCS^- , CNS^- , or CSN^- .

Structure			
Formal charge	–1 0 0	–2 +1 0	–2 +2 –1

- Note that the sum of the formal charges in each case is equal to the **charge of the ion (–1)**.
- However, the first arrangement of atoms is preferred because it **has the lowest number of atoms with nonzero formal charges** (Guideline 2).
- Also, it places **the least electronegative atom in the center**, and the negative charge on the more electronegative element (Guideline 4).

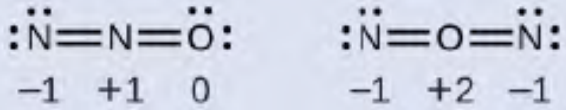
Using Formal Charge to Determine Molecular Structure

Nitrous oxide, N_2O, commonly known as laughing gas, is used as an anesthetic in minor surgeries, such as the routine extraction of wisdom teeth. Which is the likely structure for nitrous oxide?

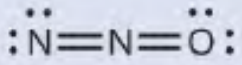


Solution

Determining formal charge yields the following:



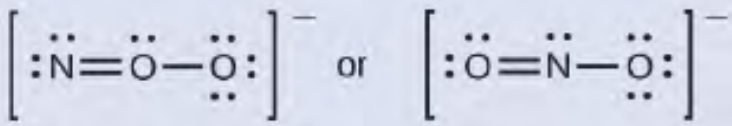
The structure with a terminal oxygen atom best satisfies the criteria for the most stable distribution of formal charge:



The number of atoms with formal charges are minimized (Guideline 2), and there is no formal charge larger than one (Guideline 2). This is again consistent with the preference for having the less electronegative atom in the central position.

Check Your Learning

Which is the most likely molecular structure for the nitrite (NO_2^-) ion?

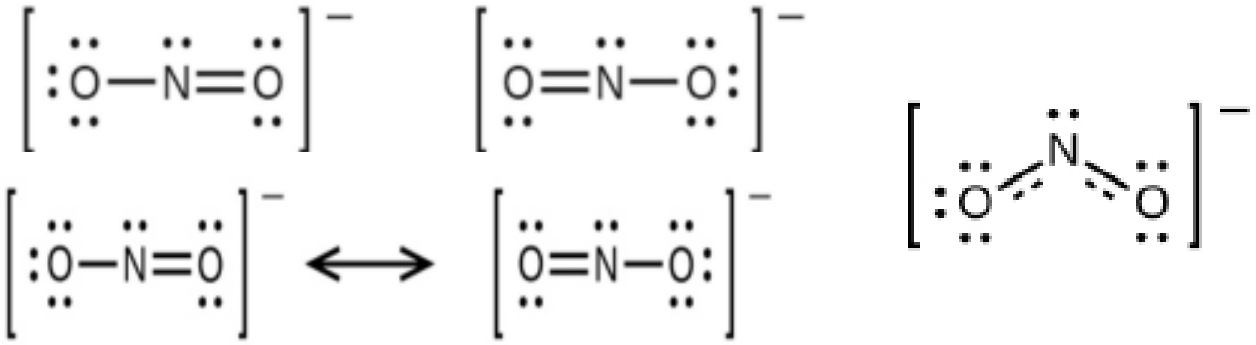


Answer: ONO^-

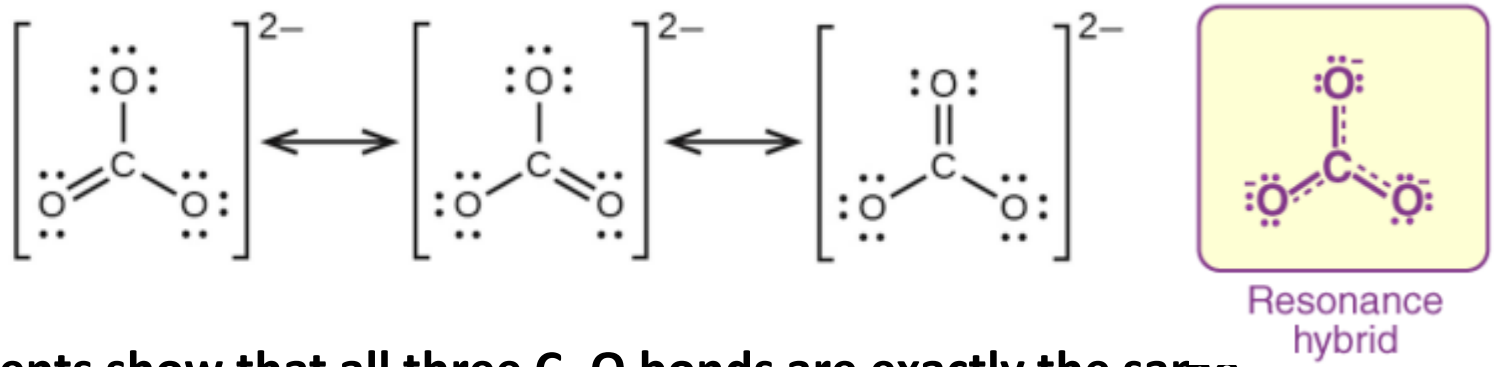
6.4.3. Resonance

Resonance refers to when **more than one valid Lewis structure** can exist for a molecule. The actual structure lies somewhere in **between the two as an average**.

- The more likely structure for the nitrite anion may actually be drawn in two different ways,



- The carbonate anion, CO_3^{2-}, provides a second example of resonance:



experiments show that all three C–O bonds are exactly the same.

6.5. Strengths of ionic and covalent bonds

- A bond's strength describes **how strongly each atom is joined to another atom**, and therefore **how much energy is required to break the bond between** the two atoms.
- An ionic compound is stable because of **the electrostatic attraction** between its positive and negative ions.
- **The lattice energy of a compound is a measure of the strength** of this attraction.

6.5.1. Ionic bond strength and lattice energy

- The lattice energy ($\Delta H_{lattice}$) of an ionic compound is defined as **the energy required to separate one mole of the solid into its component gaseous ions**.
- For the ionic solid MX, the lattice energy is the enthalpy change of the process:

$$MX\,(s) \longrightarrow M^{+}\,(g) + X^{-}\,(g) \quad \Delta H_{lattice}$$

- A **larger magnitude for lattice energy** indicates a more stable ionic compound

Example: sodium chloride, $\Delta H_{lattice}$ = +769 kJ

- it requires 769 kJ **to separate one mole of solid NaCl** into gaseous Na^+ and Cl^- ions.

The lattice energy $\Delta H_{lattice}$ of an ionic crystal can be expressed by the following equation (derived from Coulomb's law, governing the forces between electric charges):

$$\Delta Hlattice = \frac{C(Z^+)}{Z^-} R_0$$

Where **C** is a **constant that depends on the type of crystal structure**;
Z^+ and **Z^-** are the **charges on the ions**; and
Ro is the **interionic distance** (the sum of the radii of the positive and negative ions).

❑Different interatomic distances produce different lattice energies.

6.5.2. Bond strength of covalent bond

- Stable molecules exist because **covalent bonds hold** the atoms together.
- Bond strength of a covalent bond is the **measure of energy required to break it**.
  - It is, the energy necessary to **separate the bonded atoms**.
  - Separating any pair of bonded atoms requires energy.
  - The **stronger a bond, the greater the energy required** to break it.
  - The energy required to break a specific covalent bond in one mole of gaseous molecules is called the **bond energy or the bond dissociation energy**.

6.5.2. Bond strength of covalent bond

- The bond energy for a diatomic molecule, D_{X-Y} , is defined as the **standard enthalpy change for the endothermic reaction**:

$$XY(g) \rightarrow X(g) + Y(g) \qquad D_{X-Y}$$

For example, the bond energy of the pure covalent H–H bond, D_{H-H}, is 436 kJ per mole of H–H bonds broken:

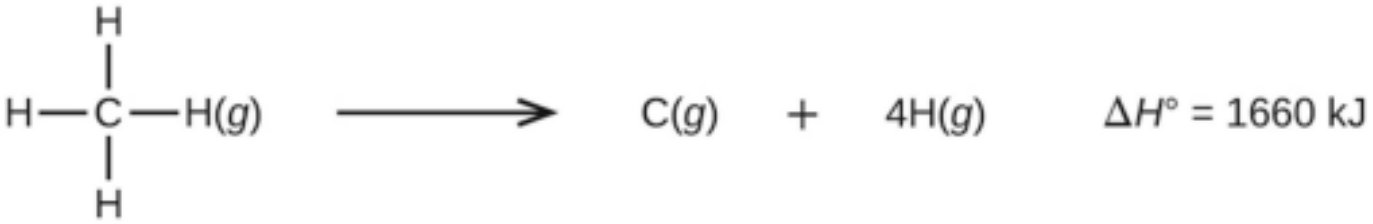
$$H_2(g) \rightarrow 2H(g) \qquad D_{H-H} = 436kJ$$

- Molecules with three or more atoms have two or more bonds.

- The **sum of all bond energies in such a molecule** is equal to the standard enthalpy change for the endothermic reaction that breaks all the bonds in the molecule.

6.5.2. Bond strength of covalent bond

Example: The sum of the four C–H bond energies in CH_4, 1660 kJ, is equal to the standard enthalpy change of the reaction:



- The **average C–H bond energy**, D_{C-H}, is 1660/4 = 415 kJ/mol because there are four moles of C– H bonds broken per mole of the reaction.
- Although the four C–H bonds are equivalent in the original molecule, **they do not each require the same energy to break; once the first bond is broken** (which requires 439 kJ/mol), the remaining bonds are easier to break.
- The **415 kJ/mol value is the average**, not the exact value required to break any one bond.

6.5.2. Bond strength of covalent bond

- The strength of a bond between two atoms increases **as the number of electron pairs in the bond increases**.
- As the **bond length decreases**, the **bond strength increases**.
- we find that **triple bonds are stronger and shorter than double bonds** between the same two atoms; likewise, **double bonds are stronger and shorter than single bonds** between the same two atoms.
- When one atom bonds to various atoms in a group, **the bond strength typically decreases as we move down the group**.

Eg. C–F is 439 kJ/mol, C–Cl is 330 kJ/mol, and C–Br is 275 kJ/mol.

Refere to Table 6.2: Bond Energies (kJ/mol) and Table 6.3: Average Bond Lengths and Bond Energies for Some Common Bonds

Bond Energies (kJ/mol)

Bond	Bond Energy	Bond	Bond Energy	Bond	Bond Energy
H–H	436	C–S	260	F–Cl	255
H–C	415	C–Cl	330	F–Br	235
H–N	390	C–Br	275	Si–Si	230
H–O	464	C–I	240	Si–P	215
H–F	569	N–N	160	Si–S	225
H–Si	395	N = N	418	Si–Cl	359
C–C	345	N–Br	245	P–Br	270
C = C	611	O–O	140	P–I	215
C ≡ C	837	O = O	498	S–S	215
C–N	290	O–F	160	S–Cl	250
C = N	615	O–Si	370	S–Br	215
C ≡ N	891	O–P	350	Cl–Cl	243
C–O	350	O–Cl	205	Cl–Br	220
C = O	741	O–I	200	Cl–I	210
C ≡ O	1080	F–F	160	Br–Br	190

Average Bond Lengths and Bond Energies for Some Common Bonds

Bond	Bond Length (Å)	Bond Energy (kJ/mol)
C–C	1.54	345
C = C	1.34	611
C ≡ C	1.20	837
C–N	1.43	290
C = N	1.38	615
C ≡ N	1.16	891
C–O	1.43	350
C = O	1.23	741
C ≡ O	1.13	1080

6.5.2. Bond strength of covalent bond

- **An exothermic reaction** (ΔH negative, heat produced) results **when the bonds in the products are stronger than the bonds in the reactants**.
- **An endothermic reaction** (ΔH positive, heat absorbed) results **when the bonds in the products are weaker than those in the reactants.**
- The **enthalpy change**, ΔH, for a chemical reaction is approximately equal to the sum of the energy required **to break all bonds in the reactants** (energy "in", positive sign) plus the energy **released when all bonds are formed in the products** (energy "out," negative sign).
- This can be expressed mathematically in the following way:

$$\Delta H = \Sigma D_{bonds\ broken} - \Sigma D_{bonds\ formed}$$

the symbol Σ means "the sum of" and D represents the bond energy in kilojoules per mole, which is always a positive number.

6.5.2. Bond strength of covalent bond

Consider the following reaction:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g) \text{ or}$$

$$H-H(g) + Cl-Cl(g) \rightarrow 2H-Cl(g)$$

To form two moles of HCl, one mole of H–H bonds and one mole of Cl–Cl bonds must be broken. The energy required to break these bonds is the sum of the bond energy of the H–H bond (436 kJ/mol) and the Cl–Cl bond (243 kJ/mol). During the reaction, two moles of H–Cl bonds are formed (bond energy = 432 kJ/mol), releasing 2 × 432 kJ; or 864 kJ. Because the bonds in the products are stronger than those in the reactants, the reaction releases more energy than it consumes:

$$\Delta H = \Sigma D_{\text{bonds broken}} - \Sigma D_{\text{bonds formed}}$$

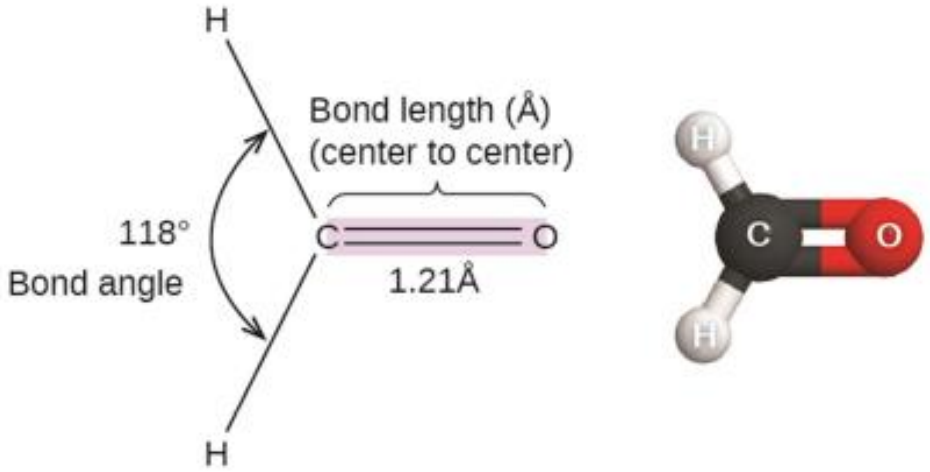
$$\Delta H = [DH - H + DCl - Cl] - 2DH -$$

$$\Delta H = [436 + 243] - 2(432) = -185 kJ$$

This excess energy is released as heat, so the reaction is exothermic.

6.5.2. Bond strength of covalent bond

- Molecular structure is actually three-dimensional, and it is important to be able to describe molecular bonds **in terms of their distances, angles, and relative arrangements in space**
- A bond angle is the **angle between any two bonds** that include a common atom, usually measured in degrees.
- A bond distance (bond length) is **the distance between the nuclei** of two bonded atoms along the straight line joining the nuclei.
- Bond distances are measured in Ångstroms (1 Å = 10^{-10} m) or picometers (1 pm = 10^{-12} m, 100 pm = 1 Å).



6.6. Molecular structure and polarity

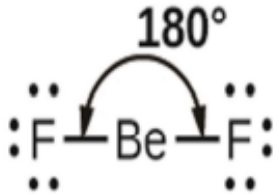
6.6.1. Valence shell electron-pair repulsion theory (VSEPR) Theory

- VSEPR theory enables us to predict the molecular structure, including approximate bond angles around a central atom, of a molecule **from an examination of the number of bonds and lone electron pairs in its Lewis structure**.
- The VSEPR model assumes that electron pairs in the valence shell of a central atom will adopt **an arrangement that minimizes repulsions between these electron pairs by maximizing the distance between them**.
- The electrons in the valence shell of a central atom form either **bonding pairs** of electrons, located primarily between bonded atoms, or **lone pairs**.
- The **electrostatic repulsion of these electrons is reduced when the various regions of high electron density assume positions** as far from each other as possible.

6.6.1. Valence shell electron-pair repulsion theory (VSEPR) Theory

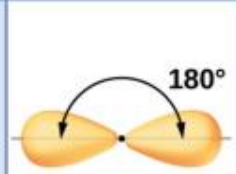
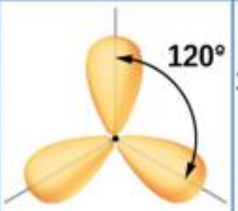
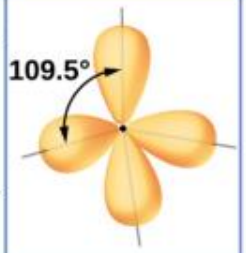
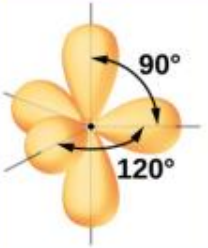
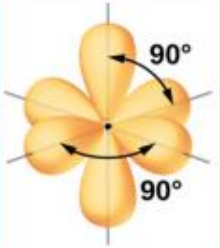
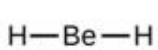
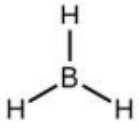
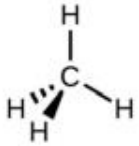
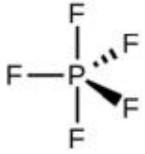
- VSEPR theory **predicts the arrangement of electron pairs around each central atom** and, usually, the correct arrangement of atoms in a molecule.

Example: The BeF_2 molecule adopts a linear structure in which the two bonds are as far apart as possible, on opposite sides of the **Be** atom.



- The basic electron-pair geometries predicted by VSEPR theory **maximize the space around any region of electron density** (bonds or lone pairs).

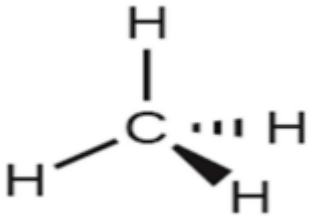
6.6.1. VSEPR Theory

Number of regions	Two regions of high electron density (bonds and/or unshared pairs)	Three regions of high electron density (bonds and/or unshared pairs)	Four regions of high electron density (bonds and/or unshared pairs)	Five regions of high electron density (bonds and/or unshared pairs)	Six regions of high electron density (bonds and/or unshared pairs)
Spatial arrangement					
Line-dash-wedge notation					
Electron pair geometry	Linear; 180° angle	Trigonal planar; all angles 120°	Tetrahedral; all angles 109.5°	Trigonal bipyramidal; angles of 90° or 120° An attached atom may be equatorial (in the plane of the triangle) or axial (above or below the plane of the triangle).	Octahedral; all angles 90° or 180°

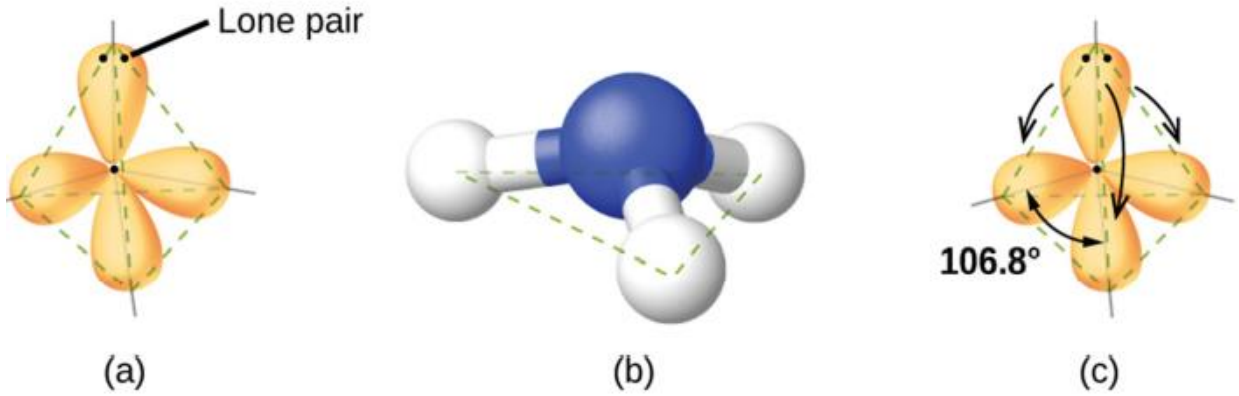
Electron-pair Geometry versus Molecular Structure

- It is important to note that electron-pair geometry around a central atom is not the same thing as its molecular structure.
- The electron-pair geometries describe all regions where electrons are located, bonds as well as lone pairs.
- Molecular structure describes the location of the atoms, not the electrons.

Methane



Ammonia



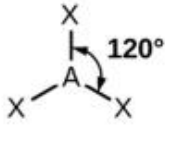
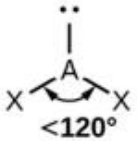
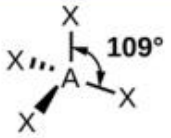
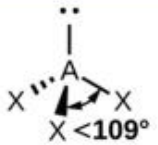
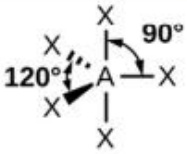
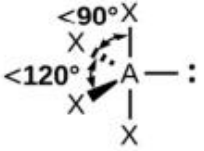
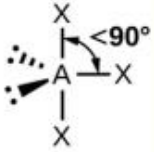
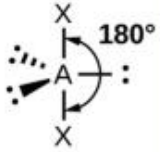
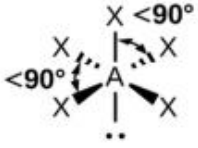
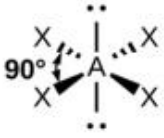
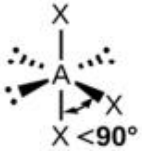
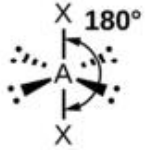
- VSEPR theory predicts distortions by establishing an order of repulsions and an order of the amount of space occupied by different kinds of electron pairs.
- The order of electron-pair repulsions from greatest to least repulsion is:

Lone pair-lone pair>lone pair-bonding pair>bonding pair-bonding pair

- A lone pair of electrons occupies a larger region of space than the electrons in a triple bond; in turn, electrons in a triple bond occupy more space than those in a double bond, and so on.
- The order of sizes from largest to smallest is:

Lone pair>triple bond>double bond>single bond

6.6.1. VSEPR Theory

Number of electron pairs	Electron pair geometries: 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal planar	 Bent or angular			
4	 Tetrahedral	 Trigonal pyramid	 Bent or angular		
5	 Trigonal bipyramid	 Sawhorse or seesaw	 T-shape	 Linear	
6	 Octahedral	 Square pyramid	 Square planar	 T-shape	 Linear

The molecular structures are identical to the electron-pair geometries when there are no lone pairs present

Predicting Electron Pair Geometry and Molecular Structure

The following procedure uses VSEPR theory to determine the electron pair geometries and the molecular structures:

1. Write the Lewis structure of the molecule or polyatomic ion.
2. Count the number of regions of electron density (lone pairs and bonds) around the central atom. A single, double, or triple bond counts as one region of electron density.
3. Identify the electron-pair geometry based on the number of regions of electron density: linear, trigonal planar, tetrahedral, trigonal bipyramidal, or octahedral
4. Use the number of lone pairs to determine the molecular structure.

✓ If more than one arrangement of lone pairs and chemical bonds is possible, choose the one that will minimize repulsions, remembering that **lone pairs occupy more space** than multiple bonds, which occupy more space than single bonds

Predicting Electron-pair Geometry and Molecular Structure: CO_2 and BCl_3

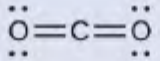
Predict the electron-pair geometry and molecular structure for each of the following:

(a) carbon dioxide, CO_2, a molecule produced by the combustion of fossil fuels

(b) boron trichloride, BCl_3, an important industrial chemical

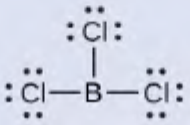
Solution

(a) We write the Lewis structure of CO_2 as:



This shows us two regions of high electron density around the carbon atom—each double bond counts as one region, and there are no lone pairs on the carbon atom. Using VSEPR theory, we predict that the two regions of electron density arrange themselves on opposite sides of the central atom with a bond angle of 180°. The electron-pair geometry and molecular structure are identical, and CO_2 molecules are linear.

(b) We write the Lewis structure of BCl_3 as:



Thus we see that BCl_3 contains three bonds, and there are no lone pairs of electrons on boron. The arrangement of three regions of high electron density gives a trigonal planar electron-pair geometry. The B–Cl bonds lie in a plane with 120° angles between them. BCl_3 also has a trigonal planar molecular structure (**Figure 7.21**).

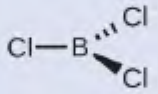


Figure 7.21

The electron-pair geometry and molecular structure of BCl_3 are both trigonal planar. Note that the VSEPR geometry indicates the correct bond angles (120°), unlike the Lewis structure shown above.

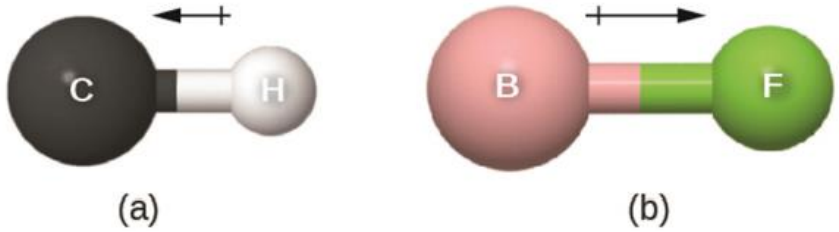
6.6.2. Molecular structure and dipole moment

- **polar covalent bonds connect two atoms with differing electronegativities**, leaving one atom with a partial positive charge (δ^+) and the other atom with a partial negative charge (δ^-).
- This separation of charge gives rise to a **bond dipole moment**.
- The magnitude of a bond dipole moment is represented by the Greek letter mu (μ)

$$\mu = Qr$$

where Q is the magnitude of the partial charges (determined by the electronegativity difference) and r is the distance between the charges

- The bond moment can be represented as a vector, a quantity having both direction and magnitude



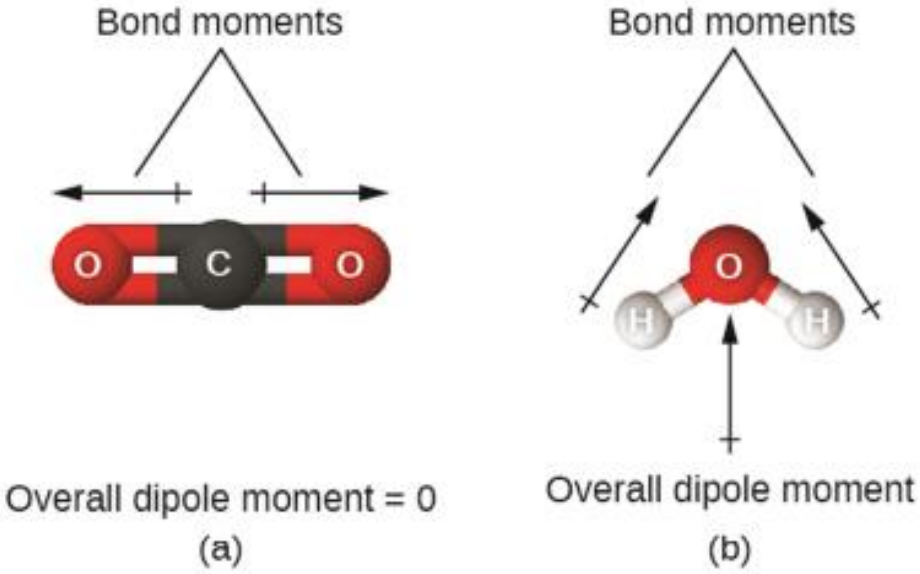
(a) (b)

(a) There is a small difference in electronegativity between C and H, represented as a short vector.

(b) The electronegativity difference between B and F is much larger, so the vector representing the bond moment is much longer.

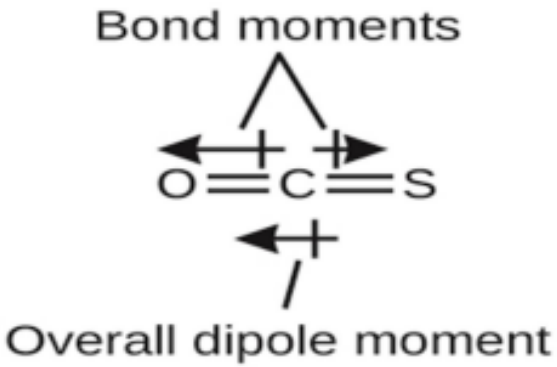
- The dipole moment measures the extent of **net charge separation in the molecule as a whol**e.

- We determine the dipole moment by **adding the bond moments in three-dimensional space**, taking into account the molecular structure.



- The overall dipole moment of a molecule depends on the individual bond dipole moments and how they are arranged.

- The OCS molecule has a structure similar to CO_2, but a sulfur atom has replaced one of the oxygen atoms.



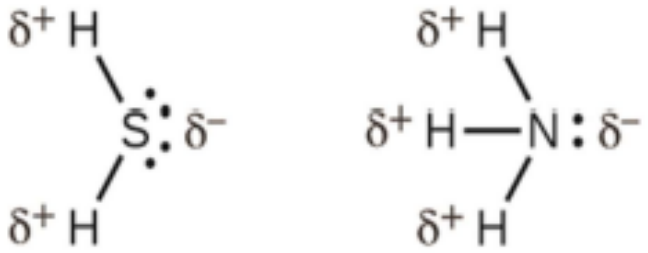
Because oxygen is more electronegative than sulfur, the oxygen end of the molecule is the negative end.



Chloromethane, CH_3Cl, is a tetrahedral molecule with three slightly polar C-H bonds and a more polar C-Cl bond.

- For molecules of high symmetry such as BF_3 (**trigonal planar**), CH_4 (**tetrahedral**), PF_5 (**trigonal bipymidal**), and SF_6 (**octahedral**), all the bonds are of identical polarity (same bond moment) and they are oriented in geometries that yield **nonpolar molecules (dipole moment is zero)**
- Molecules of less geometric symmetry, however, may be polar even when all bond moments are identical.

- Examples of such molecules include hydrogen sulfide, H_2S(**nonlinear**), and ammonia, NH_3 (**trigonal pyramidal**).



Properties of Polar Molecules

- Polar molecules tend to align when placed in an electric field with the positive end of the molecule oriented toward the negative plate and the negative end toward the positive plate

